Exam 2

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Exercise 1

Determine the null space for the matrix:

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

(state the answer in parametric form, i.e., as the general expression for the vectors belonging to the null space).

What is the dimension of the null space and the vector space to which it is a subspace? Give a basis for the null space.

✅ Exercise 1: Null Space of a Matrix

Given the matrix:

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & -3 \end{bmatrix}$$

We want to:

- Determine the **null space** (i.e., solve $A\mathbf{x} = 0$)
- Express in **parametric form**
- State the **dimension** and a **basis** for the null space
- Identify the **ambient vector space**

Step 1: Set up the equation $A\mathbf{x} = 0$

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$. The system becomes:

$$\begin{aligned} x_1 + 2x_2 - x_4 &= 0 \\ x_3 - 3x_4 &= 0 \end{aligned}$$

Step 2: Solve the system

From the second equation: $x_3 = 3x_4$

From the first: $x_1 = -2x_2 + x_4$

Let $x_2 = s$, $x_4 = t$. Then:

$$\mathbf{x} = \begin{bmatrix} -2s + t \\ s \\ 3t \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 3 \\ 1 \end{bmatrix}$$

✅ **Final Answer:**

- **Parametric form:**

✅ Final Answer:

- **Parametric form:**

$$\mathbf{x} = s\begin{bmatrix}-2\\1\\0\\0\end{bmatrix} + t\begin{bmatrix}1\\0\\3\\1\end{bmatrix}$$

- **Dimension** of null space = 2
- **Basis** = $\left\{\begin{bmatrix}-2\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\3\\1\end{bmatrix}\right\}$
- Null space is a subspace of $\mathbb{R}^4$

Exercise 2

The following matrix has the eigenvalue $\lambda = -1$:

$$\boldsymbol{A} = \begin{bmatrix} -1 & 0 & 0\\ 0 & 1 & 2\\ 0 & 2 & 1\end{bmatrix}$$

Determine the algebraic multiplicity of the eigenvalue as well as the dimension and span of the eigenspace of $\boldsymbol{A}$ corresponding to that eigenvalue; argue for the dimensionality in the form of a basis.

Argue that the set of all eigenvectors of $\boldsymbol{A}$ forms a basis for $\mathbb{R}^3$.

✅ Exercise 2: Eigenvalue Analysis

Given:

$$A = \begin{bmatrix} -1 & 0 & 0\\ 0 & 1 & 2\\ 0 & 2 & 1\end{bmatrix}$$

We are told that $\lambda = -1$ is an eigenvalue. We must:

1. Determine its **algebraic multiplicity**
2. Find the **dimension and span** of the **eigenspace**
3. **Argue that the set of all eigenvectors of A forms a basis for** $\mathbb{R}^3$

🔹 Step 1: Characteristic Polynomial

We find the eigenvalues from $\det(A - \lambda I) = 0$.

$$\det\left(\begin{bmatrix} -1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 2 \\ 0 & 2 & 1-\lambda \end{bmatrix}\right)$$

Use cofactor expansion along the first row:

$$= (-1-\lambda) \cdot \det\left(\begin{bmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{bmatrix}\right)$$

Compute the 2×2 determinant:

$$(1-\lambda)^2 - 4 = \lambda^2 - 2\lambda - 3$$

Now:

$$\det(A - \lambda I) = (-1-\lambda)(\lambda^2 - 2\lambda - 3)$$

Factor:

$$(-1-\lambda)(\lambda-3)(\lambda+1)$$

Or rearranged:

$$(\lambda+1)^2(\lambda-3)$$

✅ Eigenvalues and Algebraic Multiplicities:

- $\lambda = -1$, **algebraic multiplicity** = 2
- $\lambda = 3$, algebraic multiplicity = 1

🔹 Step 2: Eigenspace for $\lambda = -1$

Solve $(A+I)\mathbf{x} = 0$. Compute:

$$A + I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{bmatrix} \to \text{Row reduce} \Rightarrow \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

From $x_2 + x_3 = 0$, so $x_2 = -x_3$. Let $x_3 = t$, and x_1 is free.

So:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ -t \\ t \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

✅ Eigenspace for $\lambda = -1$:

- **Dimension:** 2
- **Basis:**

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

🔹 Step 3: Eigenvectors Span $\mathbb{R}^3$?

Eigenvalues:

- $\lambda = -1$: eigenspace dimension 2
- $\lambda = 3$: we compute its eigenspace next

Solve $(A - 3I)\mathbf{x} = 0$:

$$A - 3I = \begin{bmatrix} -4 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 2 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow x_1 = 0,\ x_2 = x_3$$

Let $x_3 = t$, then $\mathbf{x} = t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

✅ Eigenspace for $\lambda = 3$:

- **Dimension:** 1
- **Basis:**

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

✅ Final Argument:

Total dimension of all eigenspaces:

- 2 (for $\lambda = -1$) + 1 (for $\lambda = 3$) = 3

Since $\mathbb{R}^3$ is 3-dimensional, and we have 3 linearly independent eigenvectors, the set of all eigenvectors of A **forms a basis for** $\mathbb{R}^3$.

Exercise 3

Solve the following initial value problem:

$$\boldsymbol{y}' = \begin{bmatrix} 3 & -4 \\ 1 & -2 \end{bmatrix} \boldsymbol{y} + \begin{bmatrix} 20cos(t) \\ 0 \end{bmatrix}$$

subject to $\boldsymbol{y}(0) = [0, 8]^T$.

In the solution, state your method of approach.

✅ Exercise 3: Solve the Initial Value Problem

Given the system:

$$\mathbf{y}' = \begin{bmatrix} 3 & -4 \\ 1 & -2 \end{bmatrix} \mathbf{y} + \begin{bmatrix} 20\cos t \\ 0 \end{bmatrix}, \quad \mathbf{y}(0) = \begin{bmatrix} 0 \\ 8 \end{bmatrix}$$

🔹 Step 1: General Strategy

We solve this **nonhomogeneous linear system** using the **variation of parameters** method:

1. Find the general solution of the **homogeneous system**:

$$\mathbf{y}' = A\mathbf{y}, \quad A = \begin{bmatrix} 3 & -4 \\ 1 & -2 \end{bmatrix}$$

2. Find a **particular solution** to the nonhomogeneous system:

$$\mathbf{y}_p(t)$$

3. Combine:

$$\mathbf{y}(t) = \mathbf{y}_h(t) + \mathbf{y}_p(t)$$

4. Apply the initial condition $\mathbf{y}(0) = \begin{bmatrix} 0 \\ 8 \end{bmatrix}$

🔹 Step 2: Solve the Homogeneous System

We solve:

$$\mathbf{y}' = A\mathbf{y}$$

Find Eigenvalues

Solve $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} 3-\lambda & -4 \\ 1 & -2-\lambda \end{vmatrix} = (3-\lambda)(-2-\lambda) + 4 = \lambda^2 - \lambda - 2$$

Set equal to zero:

$$\lambda^2 - \lambda - 2 = 0 \Rightarrow \lambda = 2, -1$$

Find Eigenvectors

For $\lambda = 2$:

$$(A - 2I)\mathbf{x} = \begin{bmatrix} 1 & -4 \\ 1 & -4 \end{bmatrix} \Rightarrow x_1 = 4x_2 \Rightarrow \mathbf{v}_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

For $\lambda = -1$:

$$(A + I) = \begin{bmatrix} 4 & -4 \\ 1 & -1 \end{bmatrix} \Rightarrow x_1 = x_2 \Rightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Homogeneous Solution:

$$\mathbf{y}_h(t) = c_1 e^{2t} \begin{bmatrix} 4 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

🔹 Step 3: Find a Particular Solution

We try:

$$\mathbf{y}_p(t) = \begin{bmatrix} a(t) \\ b(t) \end{bmatrix}$$

We guess:

$$\mathbf{y}_p(t) = \mathbf{v}\cos t + \mathbf{w}\sin t$$

Differentiate:

$$\mathbf{y}_p' = -\mathbf{v}\sin t + \mathbf{w}\cos t$$

Substitute into the original equation:

$$\mathbf{y}_p' = A\mathbf{y}_p + \begin{bmatrix} 20\cos t \\ 0 \end{bmatrix}$$

Group by $\sin t$ and $\cos t$:

Coefficient of $\sin t$:

$$-\mathbf{v} = A\mathbf{w}$$

Coefficient of $\cos t$:

$$\mathbf{w} = A\mathbf{v} + \begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

This gives a system:

1. $-\mathbf{v} = A\mathbf{w}$
2. $\mathbf{w} = A\mathbf{v} + \begin{bmatrix} 20 \\ 0 \end{bmatrix}$

Substitute (2) into (1):

$$-\mathbf{v} = A(A\mathbf{v} + \begin{bmatrix} 20 \\ 0 \end{bmatrix}) \Rightarrow -\mathbf{v} = A^2\mathbf{v} + A\begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

Compute:

$$A = \begin{bmatrix} 3 & -4 \\ 1 & -2 \end{bmatrix}, \quad A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -2 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 5 & -4 \\ 1 & 0 \end{bmatrix}$$

$$A\begin{bmatrix} 20 \\ 0 \end{bmatrix} = \begin{bmatrix} 60 \\ 20 \end{bmatrix}$$

So:

$$-\mathbf{v} = A^2\mathbf{v} + \begin{bmatrix} 60 \\ 20 \end{bmatrix} \Rightarrow A^2\mathbf{v} + \mathbf{v} = -\begin{bmatrix} 60 \\ 20 \end{bmatrix} \Rightarrow (A^2 + I)\mathbf{v} = -\begin{bmatrix} 60 \\ 20 \end{bmatrix}$$

$$A^2 + I = \begin{bmatrix} 6 & -4 \\ 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 6 & -4 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -60 \\ -20 \end{bmatrix}$$

Solve:

$$6v_1 - 4v_2 = -60$$
$$v_1 + v_2 = -20 \Rightarrow v_1 = -20 - v_2$$

Substitute:

$$6(-20 - v_2) - 4v_2 = -60 \Rightarrow -120 - 6v_2 - 4v_2 = -60 \Rightarrow -10v_2 = 60 \Rightarrow v_2 = -6,\ v_1 = -14$$

$$\Rightarrow \mathbf{v} = \begin{bmatrix} -14 \\ -6 \end{bmatrix} \Rightarrow \mathbf{w} = A\mathbf{v} + \begin{bmatrix} 20 \\ 0 \end{bmatrix} = \begin{bmatrix} 3(-14) + (-4)(-6) \\ -14 + (-2)(-6) \end{bmatrix} + \begin{bmatrix} 20 \\ 0 \end{bmatrix} = \begin{bmatrix} -42 + 24 \\ -14 + 12 \end{bmatrix} + \begin{bmatrix} 20 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$$

✅ Particular Solution:

$$\mathbf{y}_p(t) = \begin{bmatrix} -14 \\ -6 \end{bmatrix}\cos t + \begin{bmatrix} 2 \\ -2 \end{bmatrix}\sin t$$

◆ **Step 4: General Solution and Apply Initial Condition**

$$\mathbf{y}(t) = \mathbf{y}_h(t) + \mathbf{y}_p(t) = c_1 e^{2t} \begin{bmatrix} 4 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -14 \\ -6 \end{bmatrix} \cos t + \begin{bmatrix} 2 \\ -2 \end{bmatrix} \sin t$$

Now apply $\mathbf{y}(0) = \begin{bmatrix} 0 \\ 8 \end{bmatrix}$:

At $t = 0$, $\cos 0 = 1$, $\sin 0 = 0$:

$$\mathbf{y}(0) = c_1 \begin{bmatrix} 4 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -14 \\ -6 \end{bmatrix} = \begin{bmatrix} 4c_1 + c_2 - 14 \\ c_1 + c_2 - 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \end{bmatrix}$$

Set up system:

$$4c_1 + c_2 = 14$$
$$c_1 + c_2 = 14 \Rightarrow c_1 = 0, c_2 = 14$$

✅ **Final Solution:**

$$\boxed{\mathbf{y}(t) = 14e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -14 \\ -6 \end{bmatrix} \cos t + \begin{bmatrix} 2 \\ -2 \end{bmatrix} \sin t}$$

Exercise 4

Calculate the output of a linear time invariant system with impulse response $h[n] = -3^n u[-n-1]$ for the input $x[n] = \left(\frac{1}{2}\right)^{n-1} u[n-1]$.

✅ **Exercise 4: Discrete-Time LTI System Output**

We are given:

- Impulse response: $h[n] = -3^n u[-n-1]$
- Input: $x[n] = \left(\frac{1}{2}\right)^{n-1} u[n-1]$

We are to compute the output $y[n]$ of the linear time-invariant (LTI) system, which is:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

◆ **Step 1: Understand the Signals**

Unit Step Functions:

- $u[n-1] = \begin{cases} 1 & n \geq 1 \\ 0 & \text{otherwise} \end{cases}$
- $u[-n-1] = \begin{cases} 1 & n \leq -1 \\ 0 & \text{otherwise} \end{cases}$

So:

- $x[n] = \left(\frac{1}{2}\right)^{n-1}$ for $n \geq 1$, 0 otherwise
- $h[n] = -3^n$ for $n \leq -1$, 0 otherwise

These are **non-overlapping** signals. One is nonzero for $n \geq 1$, the other for $n \leq -1$.

🔹 Step 2: Determine the Nonzero Range of the Convolution

We want values of n for which $x[k]$ and $h[n-k]$ are both nonzero.

- $x[k] \neq 0$ only if $k \geq 1$
- $h[n-k] \neq 0$ only if $n-k \leq -1 \Rightarrow k \geq n+1$

So both are nonzero only if:

$$k \geq \max(1, n+1)$$

But since $k \geq 1$ and $k \geq n+1$, we conclude that for the sum to be non-empty:

$$n+1 \leq k \quad \Rightarrow \quad n \leq k-1$$

And since $k \geq 1$, then $n \leq 0$

🟧 **Therefore: $y[n] \neq 0$ only when $n \leq 0$**

🔹 Step 3: Compute the Convolution for $n \leq 0$

Use:

$$y[n] = \sum_{k=\max(1,n+1)}^{\infty} \left(\frac{1}{2}\right)^{k-1} \cdot (-3)^{n-k}$$

Let's simplify this with a substitution:

Let $m = k-1$ so $k = m+1$, then:

$$x[k] = \left(\frac{1}{2}\right)^{m}, \quad h[n-k] = (-3)^{n-m-1}$$

So:

$$y[n] = \sum_{m=\max(0,n)}^{\infty} \left(\frac{1}{2}\right)^{m} \cdot (-3)^{n-m-1}$$

We factor $(-3)^{n-1}$ outside:

$$y[n] = (-3)^{n-1} \sum_{m=\max(0,n)}^{\infty} \left(\frac{1}{2} \cdot \frac{1}{-3}\right)^{m} = (-3)^{n-1} \sum_{m=\max(0,n)}^{\infty} \left(-\frac{1}{6}\right)^{m}$$

Now let's distinguish two cases.

🔹 Case 1: $n \leq 0$

Then $\max(0,n) = 0$, so:

$$y[n] = (-3)^{n-1} \sum_{m=0}^{\infty} \left(-\frac{1}{6}\right)^{m} = (-3)^{n-1} \cdot \frac{1}{1+\frac{1}{6}} = (-3)^{n-1} \cdot \frac{6}{7}$$

✅ Final Answer:

$$\boxed{y[n] = \begin{cases} \frac{6}{7} \cdot (-3)^{n-1} & n \leq 0 \\ 0 & n > 0 \end{cases}}$$

Exercise 5

Calculate the impulse response of the inverse system of:

- $h[n] = 2^{n-1}u[n-1]$
- $h[n] = 4^n u[n] - 2(4)^{n-1}u[n-1]$

🔹 Part A: Inverse System for $h[n] = 2^{n-1}u[n-1]$

This is a causal exponential response starting at $n = 1$. Let's find the Z-transform:

$$h[n] = 2^{n-1}u[n-1] = x[n-1] \text{ where } x[n] = 2^n u[n]$$

So:

$$H(z) = \mathcal{Z}\{h[n]\} = z^{-1} \cdot \sum_{n=0}^{\infty} 2^n z^{-n} = \frac{z^{-1}}{1 - 2z^{-1}} = \frac{1}{z-2}$$

Now invert the system: Find $H_{\text{inv}}(z) = \frac{1}{H(z)} = z - 2$

Now take inverse Z-transform:

$$H_{\text{inv}}(z) = z - 2 = \mathcal{Z}\{\delta[n+1]\} - 2\delta[n]$$

But to keep this causal (starting at $n = 0$), we represent:

$$H_{\text{inv}}(z) = z - 2 = \mathcal{Z}\{\delta[n+1] - 2\delta[n]\}$$

This is a non-causal system, meaning its impulse response is not zero for $n < 0$. But in practice, LTI inverse systems may be non-causal.

✅ Final Answer for System 1:

$$\boxed{h_{\text{inv}}[n] = \delta[n+1] - 2\delta[n]}$$

This is the impulse response of the inverse system: it subtracts and delays.

🔹 Part B: Inverse System for $h[n] = 4^n u[n] - 2 \cdot 4^{n-1}u[n-1]$

Step 1: Simplify the expression

We interpret:

$$h[n] = 4^n u[n] - 2 \cdot 4^{n-1}u[n-1]$$

For $n = 0$: $\quad h[0] = 4^0 - 0 = 1$

For $n \geq 1$:

$$h[n] = 4^n - 2 \cdot 4^{n-1} = 4^{n-1}(4-2) = 2 \cdot 4^{n-1}$$

So we can express:

$$h[n] = \begin{cases} 1 & n = 0 \\ 2 \cdot 4^{n-1} & n \geq 1 \\ 0 & n < 0 \end{cases}$$

Step 2: Take Z-transform

Let's define:

- $h_1[n] = 4^n u[n] \quad \Rightarrow \quad H_1(z) = \frac{1}{1-4z^{-1}}$
- $h_2[n] = 4^{n-1} u[n-1] \quad \Rightarrow \quad H_2(z) = z^{-1} \cdot \frac{1}{1-4z^{-1}}$

Then:

$$H(z) = H_1(z) - 2H_2(z) = \frac{1}{1-4z^{-1}} - \frac{2z^{-1}}{1-4z^{-1}} = \frac{1-2z^{-1}}{1-4z^{-1}} = \frac{z-2}{z-4}$$

Step 3: Invert the System

$$H_{\text{inv}}(z) = \frac{1}{H(z)} = \frac{z-4}{z-2}$$

Write as partial fractions:

$$H_{\text{inv}}(z) = \frac{z-4}{z-2} = 1 + \frac{-2}{z-2}$$

Inverse Z-transform:

- $\mathcal{Z}^{-1}\{1\} = \delta[n]$
- $\mathcal{Z}^{-1}\left\{\frac{1}{z-a}\right\} = a^n u[n]$

So:

$$h_{\text{inv}}[n] = \delta[n] - 2 \cdot 2^n u[n] = \delta[n] - 2^{n+1} u[n]$$

✅ Final Answer for System 2:

$$\boxed{h_{\text{inv}}[n] = \delta[n] - 2^{n+1} u[n]}$$

✅ Summary of Final Answers:

System	Original $h[n]$	Inverse Impulse Response $h_{\text{inv}}[n]$
1	$2^{n-1} u[n-1]$	$\delta[n+1] - 2\delta[n]$
2	$4^n u[n] - 2 \cdot 4^{n-1} u[n-1]$	$\delta[n] - 2^{n+1} u[n]$